# Ayush Vora

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## Skills

- **Languages:** Python, SQL, R, VBA, Java, C++, C#, TypeScript, JavaScript, Bash, C, Go, Rust, Kotlin, Swift, PowerShell, PHP, HTML, CSS
- **Statistics & ML:** Regression, Lasso/Ridge, Time Series (ARIMA), Random Forests, Boosting, PCA, Clustering, Cross-Validation
- **Libraries:** pandas, NumPy, scikit-learn, PyTorch, TensorFlow, Matplotlib, tidyverse, ggplot2, forecast, glmnet, BeautifulSoup
- **Data & BI:** Databricks, Snowflake, Spark, PostgreSQL, MySQL, SQLite, MongoDB, Elasticsearch, Redis, Tableau, Power BI, Excel, Salesforce
- **Tools:** Jupyter, Git, GitHub Actions, Docker, AWS, Azure, GCP, Claude API, MCP, Linux, Cloudflare Workers

## Experience

**Systems Administrator Co-op**, BC Centre for Ability — Jan 2025 – Jan 2026

- Automated archival and optimization of **Salesforce** databases with **Python** and **Selenium**, freeing **520 GB** of storage across 3 databases.
- Mass-deployed **MS Office 365** to **100+ computers** via automation scripts, cutting setup time by **85%**, and coordinated client scheduling through **OneDesk** and **Outlook** alongside live feature demonstrations.
- Onboarded **40+ staff** across **6 departments**, including setting up accounts in **AD**, managing **SharePoint** and Shared **OneDrive** permissions, and deploying Windows laptops with standardized imaging, **Fortinet FortiClient** VPN access, and asset tracking.
- Administered IT infrastructure across **Active Directory**, **Entra ID**, **Intune**, **Azure**, **Hyper-V**, and managed **4 Windows servers**, resolving **25+ tickets weekly** and supporting day-to-day systems operations.

**Toolset Software Developer / Data Analyst Co-op**, ATS Corporation — Jan 2024 – Apr 2024

- Implemented and iteratively improved MS Excel Macros using **VBA** to automate tasks and generate data visuals, saving **150+ hours** of repetitive work annually.
- Built an automated data pipeline into a **Databricks** SQL server, improving **KPI dashboard** generation efficiency and enabling **ML-based anomaly and data quality detection**.
- Fixed **12 bugs** present in the Java-based report-generator **METRIX**, improving app efficiency and increasing productivity by **33%**.

**Teaching Assistant**, University of British Columbia — Sep 2024 – May 2026

- Conducted weekly labs and tutorials for **150+ students** across introductory, intermediate, and upper-year algorithms courses.
- Received a **99.5% favourable rating** from students for clarity of explanations and approachability.

**IT Technician**, Rootnet Solutions — May 2022 – Sep 2022

- Deployed **200+ laptops and desktops** successfully to **5 Air Canada offices**, including **YYZ International Airport**.
- Served as the primary point of contact for **Air Canada YYZ Branch** for technology installations, relocation, expansions, and modifications.

## Projects

**Job Market Data Pipeline** - Python, PostgreSQL, Playwright, Claude API, Docker — GitHub: job-board-site

- Built a daily **ETL pipeline** scraping **159 company career sites** with **Python** and **Playwright** into **PostgreSQL**.
- Kept data clean with **idempotent upserts** on deterministic IDs, per-host rate limiting, and run logs that flag failed or empty scrapes.
- Scored each posting **0–100** for resume fit with an **LLM**, extracting matched and missing skills as structured output, and cached scores by a hash of resume, model, and prompt version to avoid re-scoring.

**Fuel Price Spread Forecasting** - R, forecast, tidyverse, ggplot2 — GitHub: stat-443-project

- Forecast the weekly U.S. gasoline–diesel price spread (**1,361 weeks**, 1995–2021) in a team of 4, comparing **exponential smoothing**, **ARIMA/SARIMA**, and **ARIMAX** on a **104-week** rolling holdout.
- Led **EDA** and data cleaning, finding a **2008 structural break** and using **ACF** diagnostics to show the series was non-stationary.
- Selected **ARIMA(4,1,3)**, cutting RMSE **20%** below persistence with white-noise residuals; lagged oil-price regressors added no gain.

**World University Rankings Model** - R, tidyverse, glmnet, ggplot2 — GitHub: stat-301-group-project

- Modelled what drives the overall scores of **1,799 universities** (team of 3) using **EDA** and residual diagnostics.
- Detected **multicollinearity** with **VIF**, then used **Lasso regression** with **10-fold cross-validation** on a **70/30 train-test split** to cut **9 predictors to 5** with no loss in adjusted $R^2$.

**Directed Research: Distribution Support Size** - Probability Theory, LaTeX — GitHub: moment-matching

- Wrote a paper under a UBC faculty supervisor on **moment matching** lower bounds for estimating a distribution's **support size**, plus notes on **Chernoff bounds** and **Poissonization**.

## Education

**University of British Columbia**, Bachelor of Science, Computer Science and Statistics — Sep 2021 – May 2026

- **Courses:** Machine Learning, Statistical Learning, Statistical Inference, Statistical Modelling, Databases, Algorithms, Operating Systems